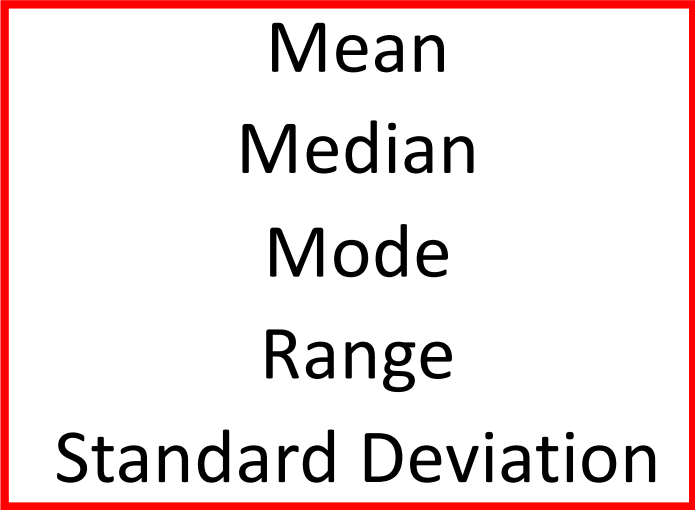


Statistics

Summarising Data

Terms used

- To describe data (or to summarise a set of scores), the following words are used:



Mean
Median
Mode
Range
Standard Deviation

Outliers

MEAN

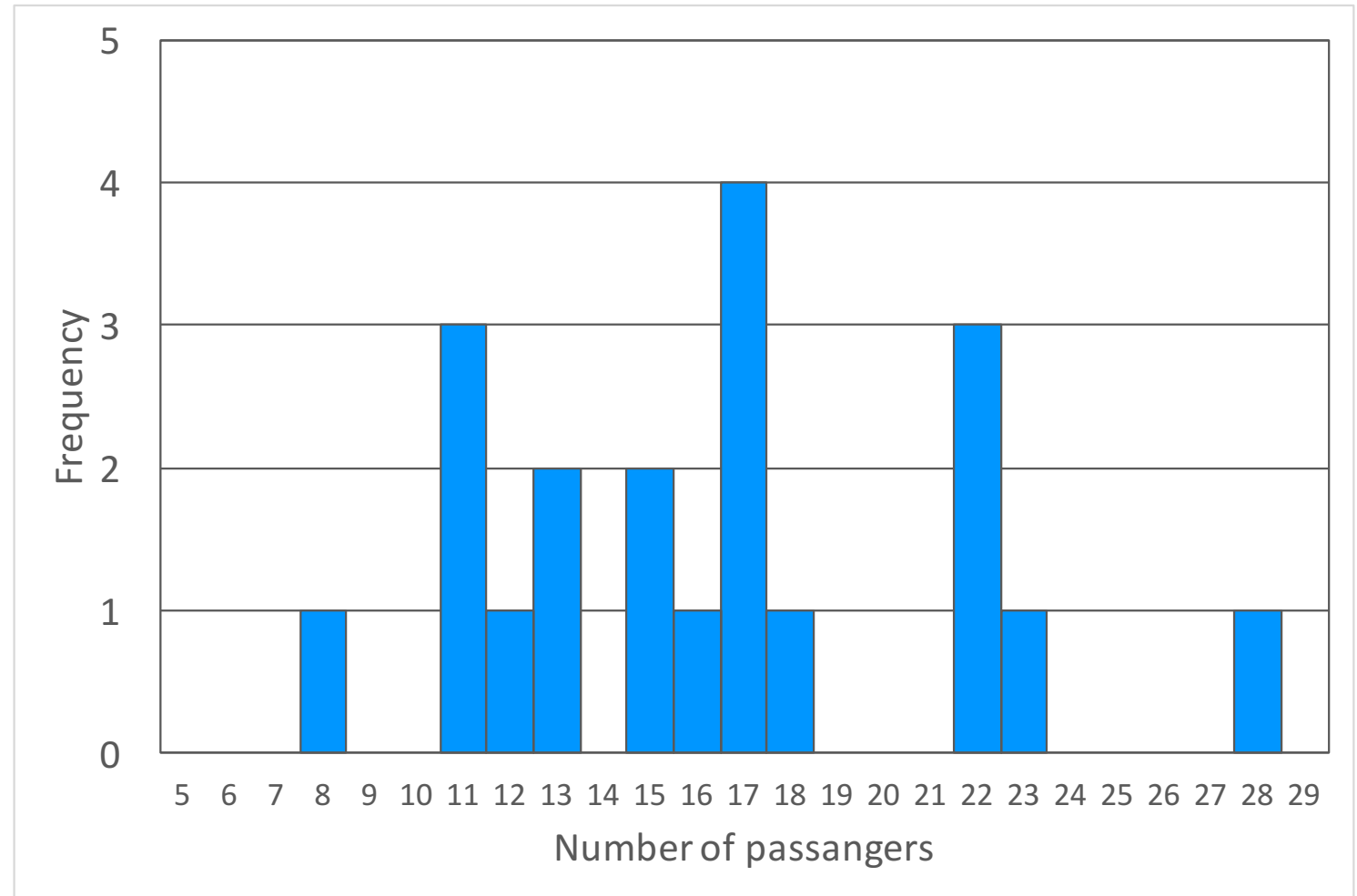
- Arithmetic average of the data (or scores)
- Measure of its centre
- Given a set of numbers:

19, 25, 29, 28, 23, 15, 27, 22, 24, 21

$$\bar{x} = \frac{19 + 25 + 29 + 28 + 23 + 15 + 27 + 22 + 24 + 21}{10} = 23.3$$

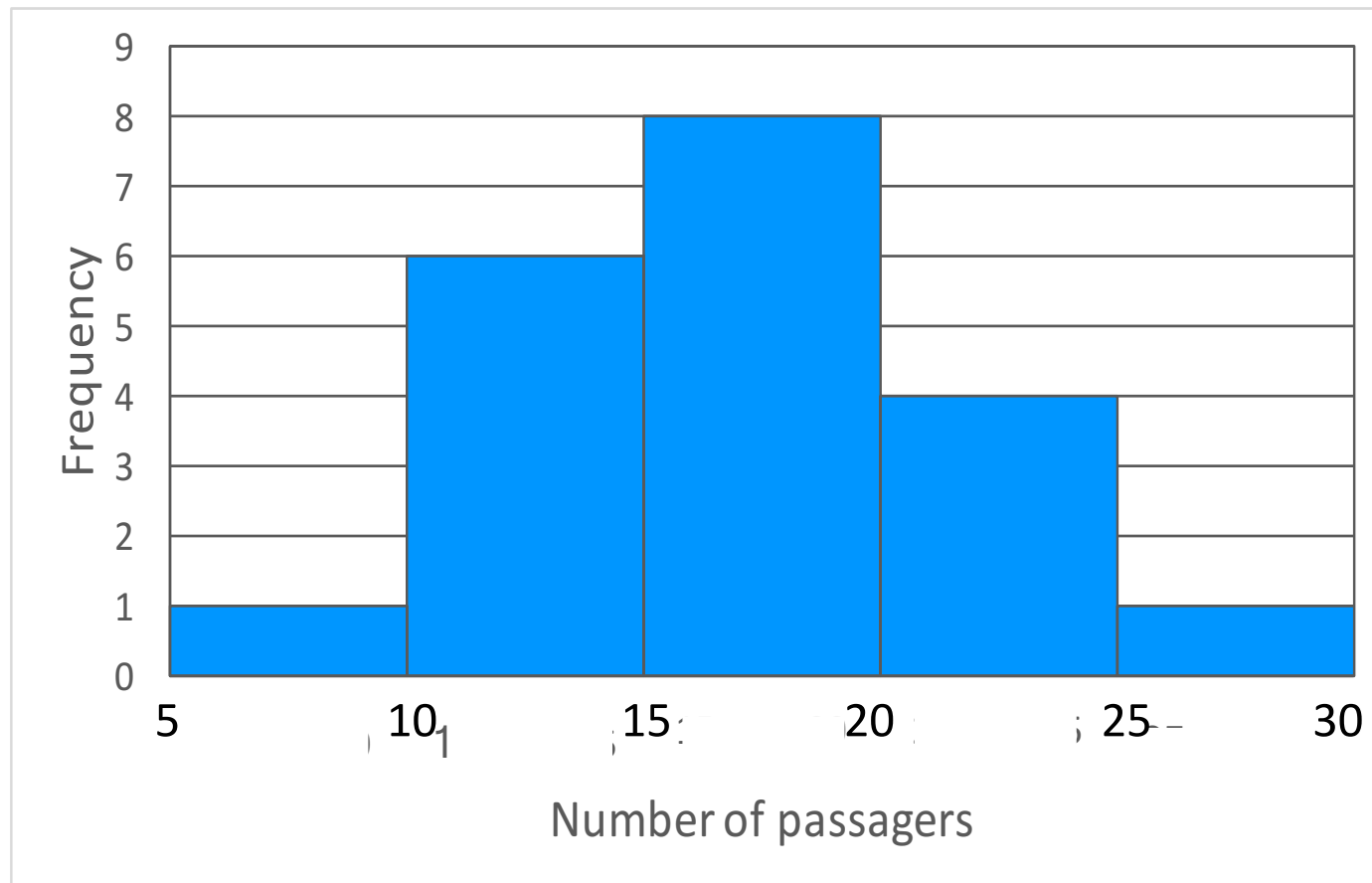
$$\text{Mean} = \frac{\text{The sum of the values}}{\text{The number of values there are}}$$

Number of passengers	Frequency
5	0
6	0
7	0
8	1
9	0
10	0
11	3
12	1
13	2
14	0
15	2
16	1
17	4
18	1
19	0
20	0
21	0
22	3
23	1
24	0
25	0
26	0
27	0
28	1
29	0



Mean = 16.4 = 17 passengers
 Median = 16.5 = 17 passengers
 Mode = 17 passengers

Number of passengers	Frequency
$5 \leq x < 10$	1
$10 \leq x < 15$	6
$15 \leq x < 20$	8
$20 \leq x < 25$	4
$25 \leq x < 30$	1



Mean = 17 passengers

Median group (median class) = $15 \leq x < 20$ passengers

Modal group (modal class) = $15 \leq x < 20$ passengers

Number of passengers	Frequency
$5 \leq x < 10$	1
$10 \leq x < 15$	6
$15 \leq x < 20$	8
$20 \leq x < 25$	4
$25 \leq x < 30$	1

Calculating mean

Use mid point of the class intervals

i.e. 7 is the mid point of $5 \leq x < 10$

$$\begin{aligned}
 \text{Mean} &= \frac{7 \times 1 + 12 \times 6 + 17 \times 8 + 22 \times 4 + 27 \times 1}{20} \\
 &= 16.5 \\
 &= 17 \text{ passengers}
 \end{aligned}$$

MEDIAN

- The middle value in an **ordered** set of values.

- Example:

23, 24, 23, 25, 23, 26, 27, 28, 28, 23, 28, 22, 26, 28, 25

Reorder:

22, 23, 23, 23, 23, 24, 25, 25, 26, 26, 27, 28, 28, 28, 28

MEDIAN = 25

Odd number of values, the median is the **middle** value.

MEDIAN

Example:

23, 24, 23, 25, 23, 26, 27, 28, 28, 23, 28, 22, 26, 28, 25, 26

Reorder:

22, 23, 23, 23, 23, 24, 25, 25, 26, 26, 26, 27, 28, 28, 28, 28

$$\text{MEDIAN} = \frac{25+26}{2} = 25.5$$

Even number of values, the median is the **average of the two middle** value.

MODE

- Mode is the value(s) that appears the most in the set of data.
- Example:

23, 24, 22, 25, 23, 26, 27, 28, 28, 23, 28, 22, 26, 28, 98

Rearrange:

22, 22, 23, 23, 23, 24, 25, 26, 26, 27, 28, 28, 28, 28, 98

Mode = 28

MODE

- Example:

23, 24, 23, 25, 23, 26, 27, 28, 28, 23, 28, 22, 26, 28, 98

Rearrange:

22, 23, 23, 23, 23, 24, 25, 26, 26, 27, 28, 28, 28, 28, 98

Bimodal, modes = 23 and 28

RANGE

- Range is the difference between the highest and lowest score.

Example:

23, 24, 23, 25, 23, 26, 27, 28, 28, 23, 28, 22, 26, 28, 25, 26

Highest = 28

lowest = 22

$\text{RANGE} = 28 - 22 = 6$

Example 1

The fifty scores shown below

14	17	15	17	12	16	19	16	17	16
16	10	15	17	18	17	14	16	17	16
18	19	20	14	15	18	18	18	15	17
15	16	17	18	16	16	13	15	18	15
17	17	15	16	19	15	17	18	13	14

Determine the mean, mode and median.

Score	10	11	12	13	14	15	16	17	18	19	20
Frequency	1	0	1	2	4	9	10	11	8	3	1

$$\begin{aligned} & 1 \times 10 + 1 \times 12 + 2 \times 13 + 4 \times 14 + 9 \times 15 + 10 \times 16 + 11 \times 17 + 8 \times 18 + 3 \times 19 + 1 \times 20 \\ = & 10 + 12 + 26 + 56 + 135 + 160 + 187 + 144 + 57 + 20 \\ = & 807 \end{aligned}$$

Hence the mean will be given by $\frac{807}{50}$ i.e. 16.14.

Example 3

Determine the mode, median, mean and range for the set of twenty scores shown in the stem and leaf diagram on the right.

12		7				
13		9	3	1		
14		2	0	3	2	7
15		2	5	5	1	9
16		8	3	8		
17		3	0			

12		7				
13		1	3	9		
14		0	2	2	3	7
15		1	2	5	5	5
16		3	8	8		
17		0	3			

Example 4

The mean of six scores is 23.5. If five of the scores were 17, 20, 19, 25 and 30 find the sixth score.

$$\frac{17 + 20 + 19 + 25 + 30 + x}{6} = 23.5$$

$$111 + x = 23.5 \times 6$$

$$x = 30$$

The sixth score is 30

Example 5

To pass a particular course a student needs to gain a mean of at least 55% in the five tests that form the course assessment. In the first four tests the student achieves marks of 46%, 57%, 54% and 57%. What percentage mark must the student gain in test five if they are to pass the course?

$$\frac{46\% + 51\% + 54\% + 57\% + x\%}{5} = 55\%$$

$$214\% + x\% = 5 \times 55\%$$

$$x\% = 61\%$$

The student needs to score at least 61% to pass the course.